

Engineering Curves: Construction of Hypocycloid

Rolling Circle Method with Tangent & Normal

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Problem Statement

Problem

Draw a hypocycloid of a circle of diameter 50 mm, which rolls inside a circular arc of radius 85 mm for one revolution. Also, draw a tangent and a normal to the epicycloid at a point 50 mm from the centre of the directing circle.

Calculated Data

$$R = 85 \text{ mm},$$

$$OC_0 = R - r = 85 - 25 = 60 \text{ mm},$$

$$\angle AOB = 360^\circ \frac{r}{R} = 360^\circ \frac{25}{85} = 105.882353^\circ,$$

$$\Delta\theta = \frac{105.882353^\circ}{12} = 8.823529^\circ,$$

$$r = \frac{50}{2} = 25 \text{ mm},$$

$$AC_0 = 25 \text{ mm},$$

Generating-circle division = 30° .

Step 1: Draw OA

$OA = 85 \text{ mm.}$

